



LaVR: Scene Latent Conditioned Generative Video Trajectory Re-Rendering using Large 4D Reconstruction Models

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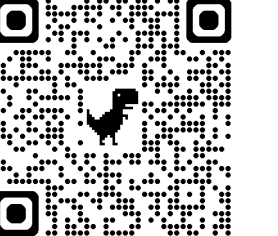
From a monocular video of a **dynamic scene**, generate novel views along a target camera trajectory.



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Let's Chat!

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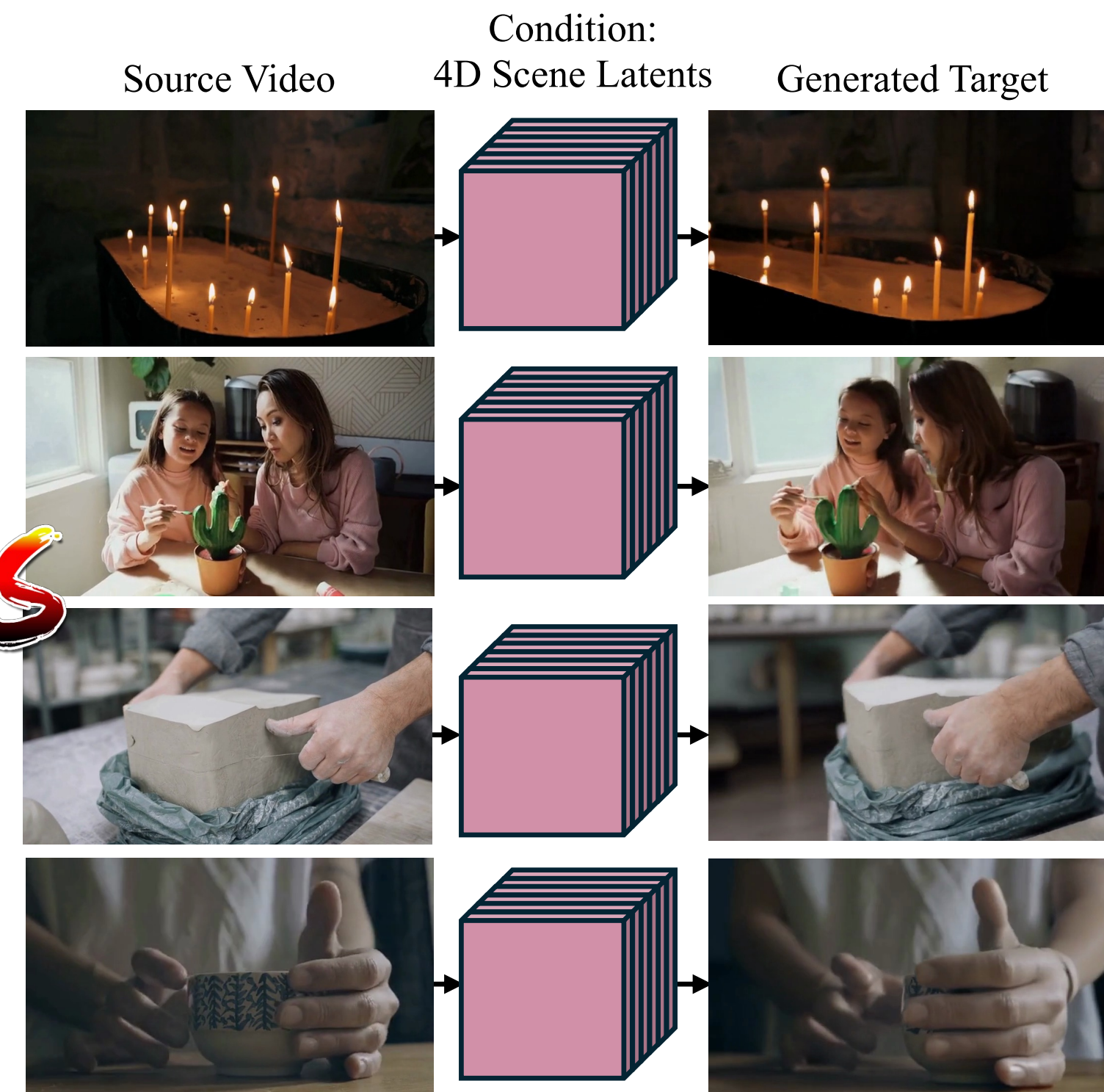
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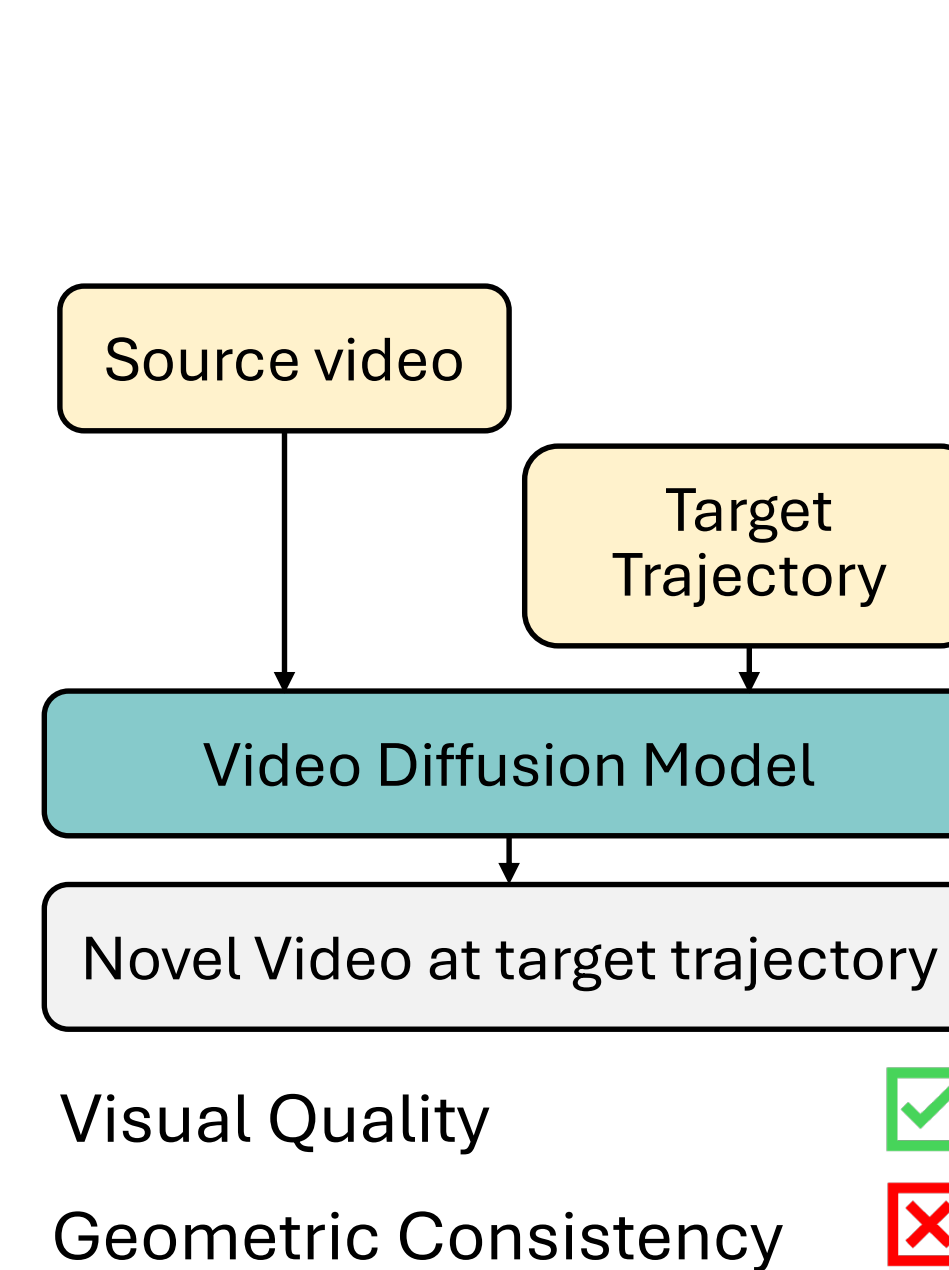
Point Cloud Based Methods



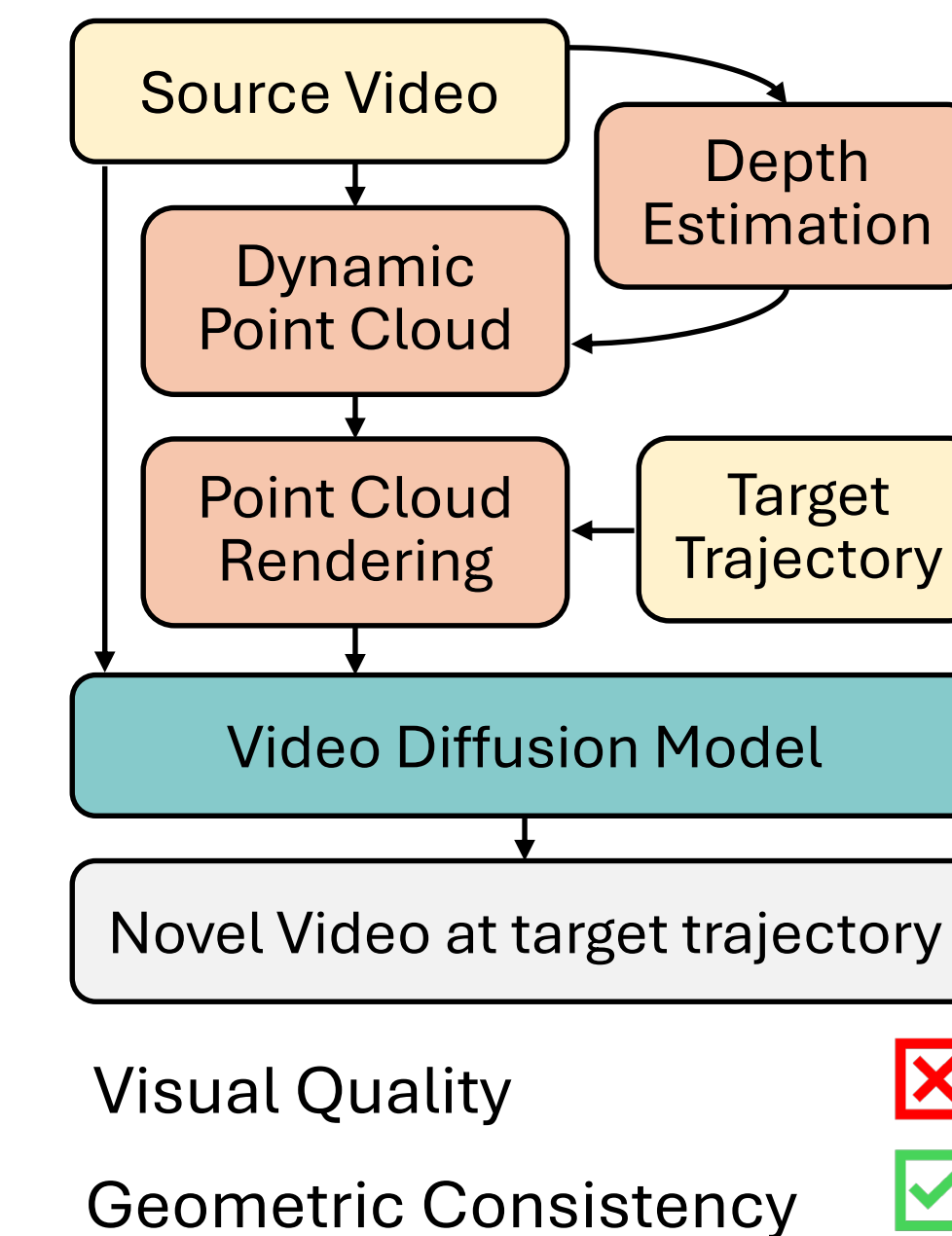
Ours



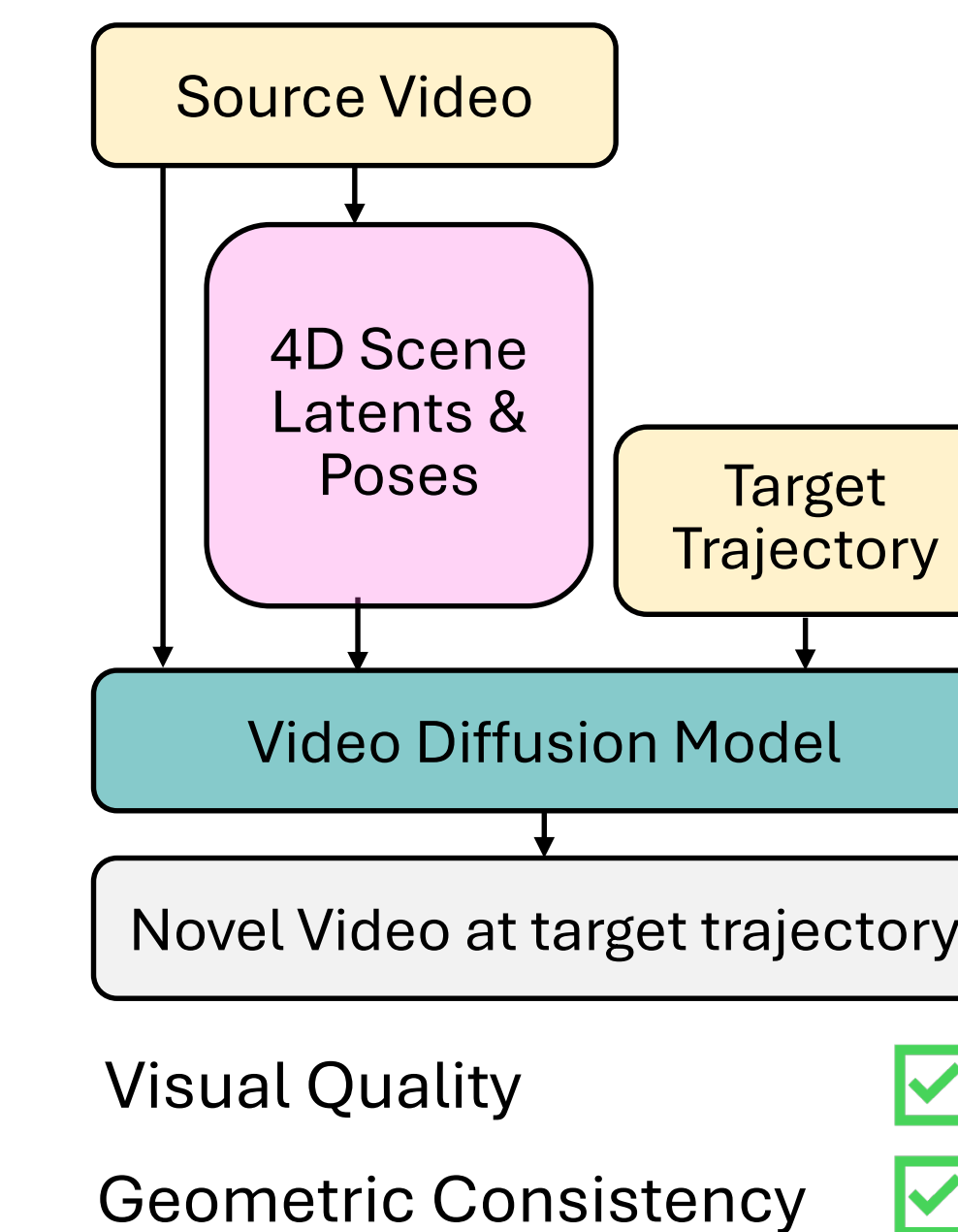
(a) Not 3D Conditioned



(b) Point Cloud Conditioned



(c) 4D Scene Latent Conditioned (Our Approach)

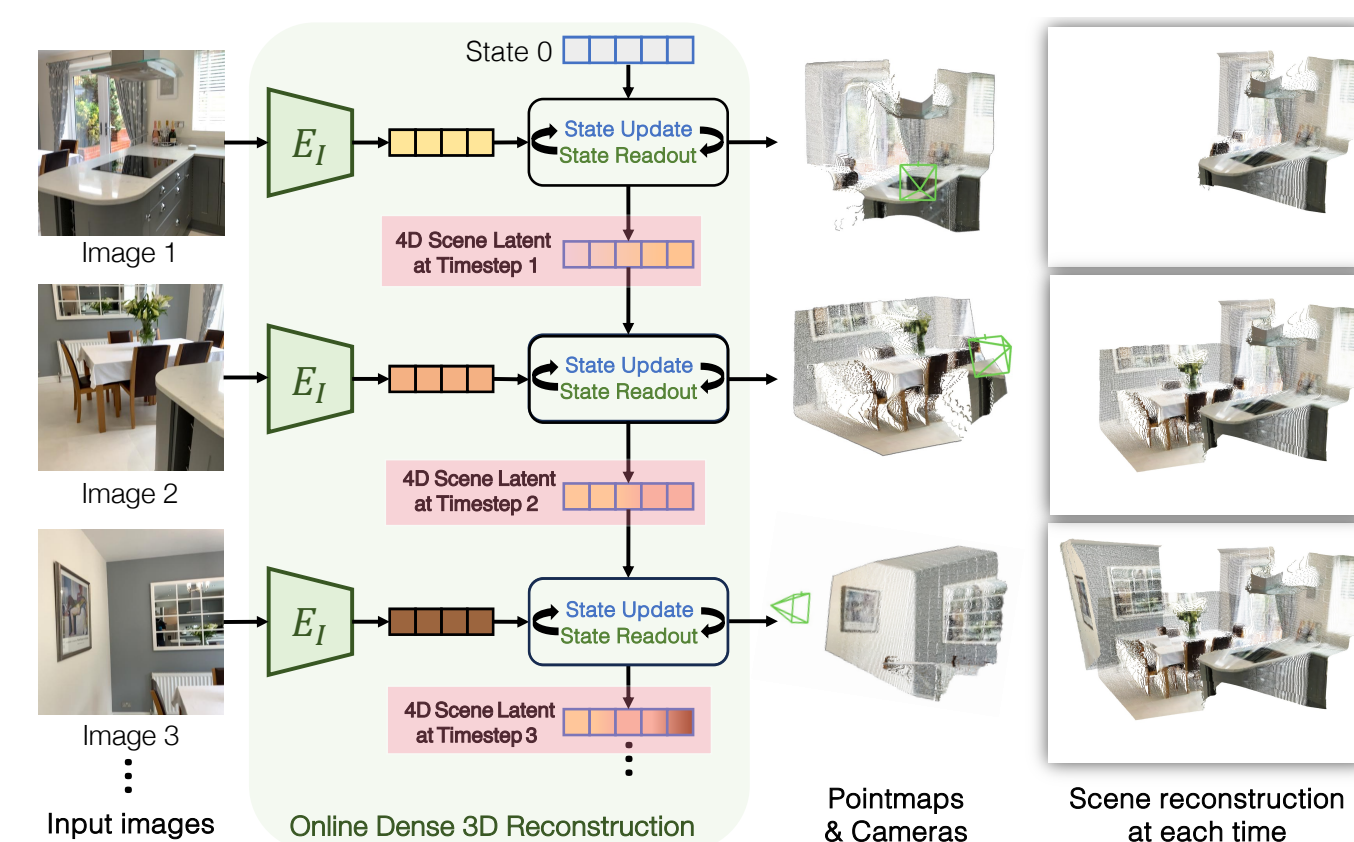


Our model is conditioned on scene latents from a large 4D reconstruction model.

Our Motivations

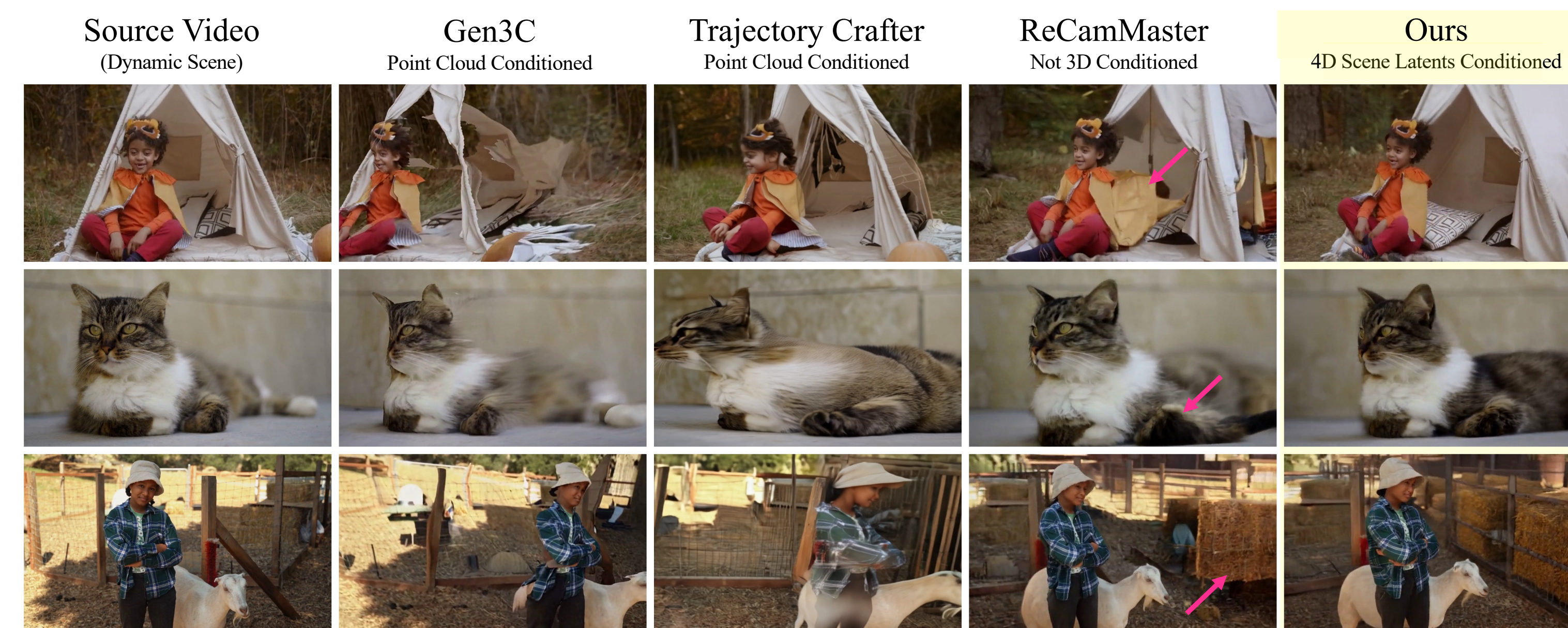
- Prior works guide video generation with re-rendered point clouds, which imposes an explicit and relatively hard geometric condition.
- However, even small depth-level errors in point clouds can appear as severe geometric artifacts under novel viewpoints, which then propagate into video generation.
- We instead use 4D scene latents from a Large 4D Reconstruction Model as a softer and more abstract geometric condition for video generation.
- This lets the pretrained video diffusion prior regularize imperfect geometry/depths into more realistic outputs.

What is a Large 4D Reconstruction Model?

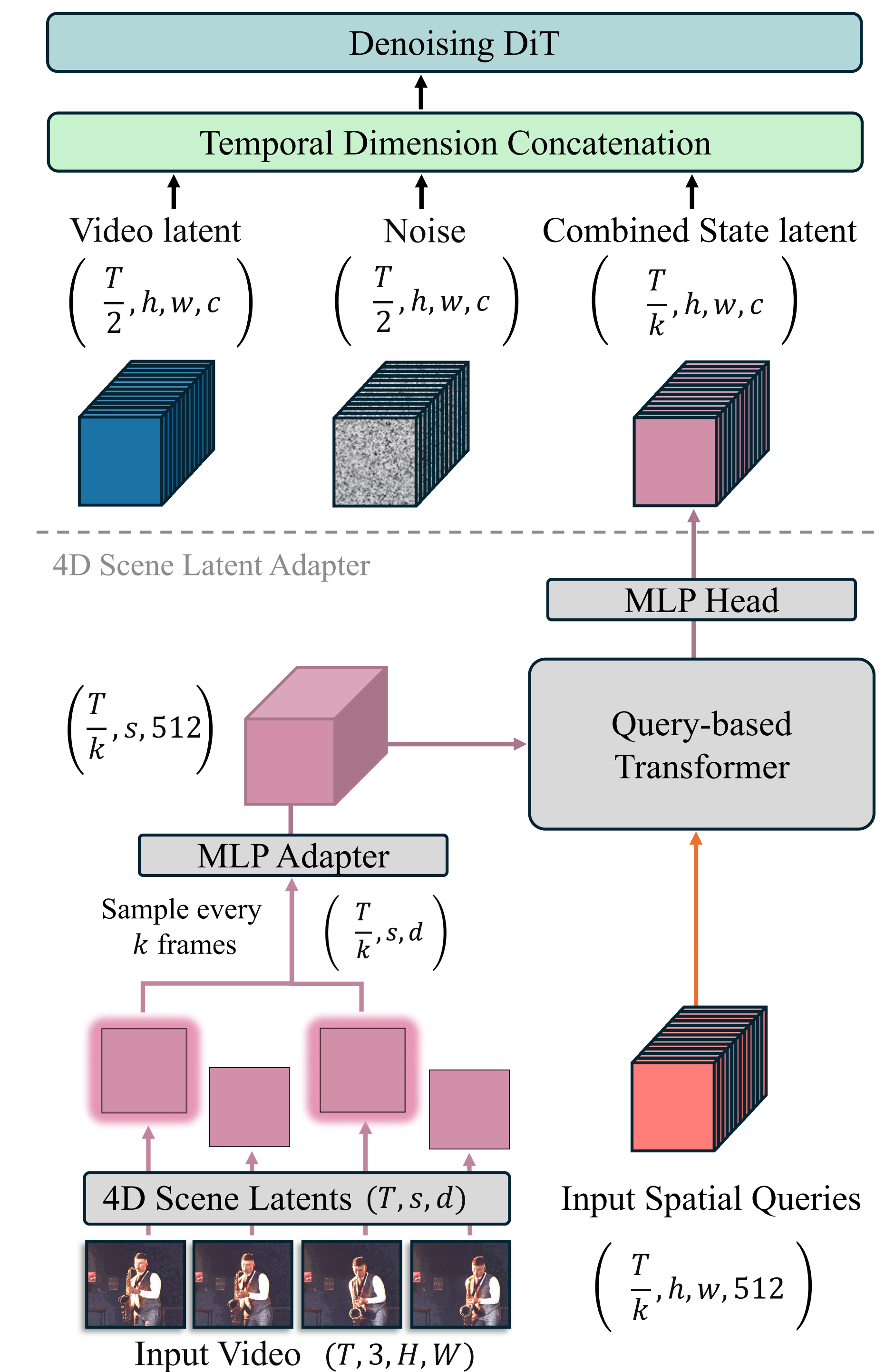


Such a model predicts the latent representation of a 4D scene at every timestep, from which it further predicts the point maps and camera poses.

*Model architecture of CUT3R. Figure by Wang Qianqian et al. CVPR 2025. More at their webpage <https://cut3r.github.io/>.



Compared to ours, point cloud based methods are prone to depth estimation errors, methods with no 3D conditioning are prone to shape deformation.



Details of Our 4D Scene Latent Conditioning.